

```
#include <iostream>
```

```
int main() {
```

```
    int choice;
```

```
    double Np, Ns, Vp, Vs;
```

```
    std::cout << "--- Transformer Calculations Menu ---\n";
```

```
    std::cout << "1. Calculate turns ratio\n2. Calculate secondary voltage\n3. Calculate primary voltage\n";
```

```
    std::cout << "Enter Choice: ";
```

```
    std::cin >> choice;
```

```
    switch(choice) {
```

```
        case 1:
```

```
            std::cout << "Enter Primary Turns: "; std::cin >> Np;
```

```
            std::cout << "Enter Secondary Turns: "; std::cin >> Ns;
```

```
            std::cout << "Turns Ratio = " << (Np / Ns) << std::endl;
```

```
            break;
```

```
        case 2:
```

```
            std::cout << "Enter Primary Voltage (V): "; std::cin >> Vp;
```

```
            std::cout << "Enter Secondary Turns: "; std::cin >> Ns;
```

```
            std::cout << "Enter Primary Turns: "; std::cin >> Np;
```

```
            std::cout << "Secondary Voltage = " << (Vp * Ns / Np) << " V" << std::endl;
```

```
            break;
```

```
        case 3:
```

```
            std::cout << "Enter Secondary Voltage (V): "; std::cin >> Vs;
```

```
            std::cout << "Enter Primary Turns: "; std::cin >> Np;
```

```
            std::cout << "Enter Secondary Turns: "; std::cin >> Ns;
```

```
            std::cout << "Primary Voltage = " << (Vs * Np / Ns) << " V" << std::endl;
```

```
            break;
```

```
        default:
```

```
            std::cout << "Invalid operational parameters configuration selected." << std::
```

```
endl;
```

```
    break;  
}  
return 0;  
}
```